

Instructor: Date:

Learner: Regional Branch:

Program: Center:

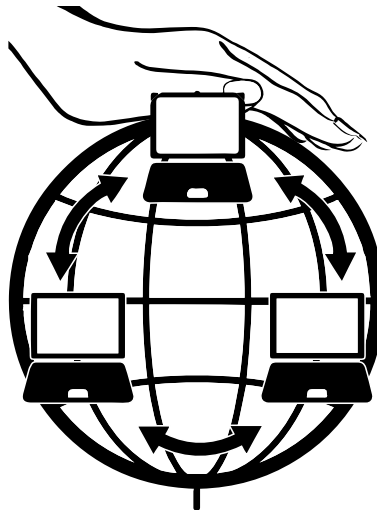
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Let's Start!

Do you know how to integrate software applications?



Learning Outcome:

Understand key information about the concepts, principles, and methodologies in simple spoken and written situations. Talk about programming techniques, use cases, kinds, and sequences in simple spoken and written situations.

Ask your partner:

- What is a common example of interactivity on the web?
- How do you usually add share functionalities to a web application?

Can Do Statements:

- I can explain what an API is.
- I can ask questions to obtain information related to API.
- I can identify important information related to API.



Let's Explore!

Jeff is catching up with a programming class.



Instructions: Read the text below.

Jeff: Hey Tina! Can you help me catch up with Mr. Kapoor class? I wasn't there yesterday.

Tina: Sure, no problem. It was quite interesting, he talked about APIs.

Jeff: Sorry, what was that again?

Tina: APIs, Application Program Interfaces. You know, all the routines, protocols, tools or methods that are used to establish communication among software components when we are programming a graphical user interface. For instance, we can use API's to set share functionalities or add comment boxes.

Jeff: So, what you are saying is that APIs allows interaction between software applications. Am I right?.

Tina: That's correct! Every time you need to access from your own application to other services, you use APIs. These interfaces are the most common tools software developers use to integrate application programs, databases, services, and devices. Before it was necessary to use high programming level for being able to display text or images on the screen, now it is easier to create software with APIs since they provide a whole set of functions to customize features with less complicated processes.

Jeff: OK. I see. Can you explain to me how an API works?

Tina: Well, we can say that an API is a kind of messenger. It is in charge of taking a user request to a specific system and then it takes it back to the user. In this way, when we are coding APIs, what we are basically doing is requesting other systems to perform a task.

Jeff: What are actual examples of the use of APIs?

Tina: We can often see how API's work, for example, when we request to read disk files in operating systems; when we access to airlines data to book flights; or even when we post videos, photos or comments in our social networks profiles.

Jeff: What type of programming language do you use to code APIs?

Tina: We usually use JSON or XML...





Let's Practice!

Can you help me catch up with Mr. Kapoor class? I wasn't there yesterday.

Instructions: Use the words from the box to correctly complete the sentences.

Word Bank

Are the most common / are all the routines / it is
easier / APIs are tools / Before it was

1. APIs, Application Program Interfaces _____, protocols, tools or methods that we used to establish communication among software components.
2. _____ that allow interaction between software applications.
3. These interfaces _____ tools software developers use to integrate application programs, databases, services, and devices.
4. _____ necessary to use high programming level for being able to display text or images on the screen, now it isn't.
5. Now _____ to create software with APIs since they provide a whole set of functions to customize features with less complicated processes.



Let's Practice!

What are actual examples of the use of APIs?

Instructions: Select the correct answer. You have 2 attempts to submit your answers.

1. I'm _____ to read a disk file, this is an example of the use of API.
a. request
b. requests
c. requesting
2. He _____ access to airlines data to book a flight.
a. having
b. is having
c. are having
3. We _____ some photos on our profile.
a. posting
b. is posting
c. are posting
4. I _____ JSON to code APIs
a. am using
b. is using
c. are using
5. They _____ that posts on social networks work with APIs.
a. am reading
b. is reading
c. are reading



Extension Activity

What are the advantages of using APIs?

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about the common uses of APIs.
- Identify key information and take notes.
- Design a brochure to explain what the advantages of using APIs in software development are.
- Share the brochure with your class.

Self-Assessment

Yes No

- I look for more information about the common uses of APIs. ☐ ☐
- I identify key information and take notes. ☐ ☐
- I design a brochure to explain what the advantages of using APIs in software development are. ☐ ☐
- I share the brochure with my class. ☐ ☐



Target Words

- routine** (noun): a section of source code that performs a specific task in the computer program
e.g. You know, all the routines, protocols, tools or methods that are useful to establish communication among software components when we are programming a graphical user interface.
- graphical user interface** (noun): a mechanism that allows the user to visualize and interact with all the information, graphical content and commands of a computer software
e.g. You know, all the routines, protocols, tools or methods that are useful to establish communication among software components when we are programming a graphical user interface.
- application program** (noun): a program that performs specific tasks according to users' needs or expectations
e.g. These interfaces are the most common tools software developers use to integrate application programs, databases, services and devices.
- customize** (verb): to adapt something according to user demands
e.g. Now it is easier to create software with APIs since they provide a whole set of functions to customize features with less complicated processes.
- JSON** (acronym): Java Script Object Notation, a programming language to interchange data
e.g. We usually use JSON or XML.



Learning Points

Simple Present and Simple Past of Verb to Be

You use the verb **be** in **present tense** or **past tense** to describe roles, qualities, states or locations.

In the **simple present**, you use three irregular forms with the **verb to be** depending on the subject:

- *I **am** aware of the importance of APIs nowadays.*
- *He **is** very talented when dealing with program applications.*
- *They **are** amazed by all the uses APIs have.*

In **simple past**, the **verb to be** has two irregular forms. The conjugation changes according to the pronouns too, we use **was** with singular pronouns and **were** with plural pronouns:

- *I **was** aware of the importance of APIs nowadays.*
- *He **was** very talented when dealing with program applications.*
- *They **were** amazed by all the uses APIs have.*

To form negative sentences using the **verb to be** in the present or past, you only have to add the word **not** after the verb to be or you can use the contracted forms **isn't/aren't** or **wasn't / weren't**.

- *He **isn't** very talented when dealing with program applications.*
- *He **wasn't** very talented when dealing with program applications.*

To make questions, the verb **be** and the pronoun switch places:

- ***Are** they amazed by all the uses APIs have?*
- ***Were** they amazed by all the uses APIs have?*

You can also use *WH-words* when you are asking for specific information:

- ***What is** the main function of API's?*
- ***What was** the main function of API's?*

Look at the chart to identify the **verb to be** forms in present and past tenses:

Verb To Be		
Pronoun	Present Tense	Past Tense
I	am	was
You	are	were
He / she / it	is	was
We	are	were
They	are	were





Learning Points

Examples

- *Hey Tim, Can you help me catch up with Mr. Kapoor class? I wasn't there yesterday.*
- *It is in charge of taking a user request to a specific system and then it takes it back to the user.*
- *Before it was necessary to use high programming level for being able to display text or images on the screen, now it is easier to create software with APIs.*





Learning Points

Present Continuous for Temporary Situations

You use the **present continuous tense** to talk about temporary actions that are taking place at the moment or in an unfinished period of time.

When you make sentences in present continuous, you use the verb to **be** as an auxiliary and you add **-ING** to the main actions:

At the moment, **I'm programming** an API to link my app with a social network.

When you **are buying** products online, you use API's to look for databases online.

To clarify the period of time, you can use time expressions such as *now, at the moment, currently, this year, nowadays* etc.

These are the structures that we can use with the **present continuous**:

Affirmative

Subject + **verb Be + main verb in ING + complement.**

Example:

He **is programming** an API to link his app with a social network.

Negative

Subject + **verb Be in negative form + main verb in ING + complement.**

Example:

He **isn't programming** an API to link his app with a social network.

Interrogative

Verb Be + subject + main verb in ING + complement?

Example:

Is he programming an API to link his app with a social network?

Examples

- *So, what you are saying is that APIs allows interaction between software applications. Am I right?*
- *In this way, when we are coding APIs, what we are basically doing is requesting other systems to perform a task.*

